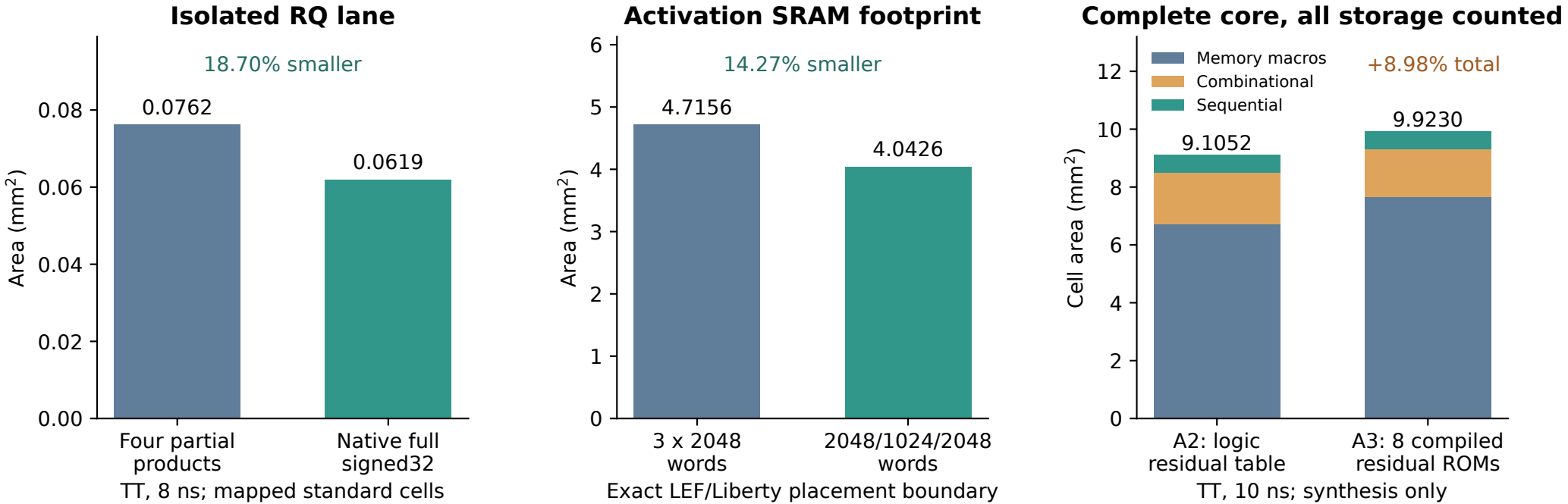
D4 / SMIC 180 nm: measured implementation trade-offs
Isolated RQ lane
18.70% smaller
0.0762
0.0619
Area (mm²)
0.00
0.02
0.04
0.06
0.08
Four partial products
Native full signed32
TT, 8 ns; mapped standard cells
Activation SRAM footprint
14.27% smaller
4.7156
4.0426
Area (mm²)
0
1
2
3
4
5
6
3 x 2048 words
2048/1024/2048 words
Exact LEF/Liberty placement boundary
Complete core, all storage counted
Memory macros
Combinational
Sequential
+8.98% total
9.1052
9.9230
Cell area (mm²)
0
2
4
6
8
10
12
A2: logic residual table
A3: 8 compiled residual ROMs
TT, 10 ns; synthesis only
Same frozen network and integer semantics. Separate area scopes; no power or post-route claim.
A2 numeric mapping reports retained; its mapped checkpoint was not saved. A3 checkpoint and report recovery verified.